

Wave Optics

When light behaves as a wave: interference, thin films, diffraction, the limits of resolution, and polarisation.

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Geometrical optics treats light as rays. That works well when every opening and obstacle is much larger than the wavelength. When light passes through openings comparable to its wavelength, or when two light waves overlap, its **wave nature** shows. Light bends around edges (**diffraction**), adds and cancels (**interference**), and vibrates in a particular direction (**polarisation**). These effects explain the colours of soap bubbles and butterfly wings. They also set the ultimate limit on how fine a detail any microscope, telescope or eye can resolve. They underpin some of the most precise measuring tools ever built, from X-ray crystallography (which revealed the structure of DNA) to optical coherence tomography, which images the layers of the retina with micrometre detail.

YOU WILL BE ABLE TO

- Use Huygens' principle, and explain the conditions for observing interference (coherence).
- Analyse Young's double-slit experiment: fringe positions and the intensity pattern.
- Explain thin-film interference, including phase changes on reflection and anti-reflection coatings.
- Describe the Michelson interferometer and optical coherence tomography.
- Analyse single-slit diffraction and the Rayleigh criterion for resolution, including the resolution limits of the eye and the microscope.
- Use diffraction gratings and Bragg's law for X-ray diffraction.
- Describe polarisation: Malus's law, Brewster's angle, and applications.

THINK FIRST

Why can't even the best light microscope show you a virus, however many lenses you stack together?

Because of **diffraction**. Light passing through a lens of finite size spreads out, so every point of the object is imaged as a small blur rather than a perfect point. The smallest separation a microscope can resolve is about $\lambda/(2NA)$, roughly 200 nm for visible light even with the best oil-immersion lenses (Example 23.5). Most viruses are 20–200 nm across, so they lie at or below this limit. To see them we need shorter wavelengths, such as the electron wavelengths of an electron microscope (Chapter 24), or ingenious “super-resolution” fluorescence techniques that won the 2014 Nobel Prize in Chemistry.

§23.1

Huygens' principle

Huygens' principle (1678) says that every point on a wavefront acts as a source of secondary spherical wavelets, and the new wavefront is the surface tangent to these wavelets a short time later. It explains rectilinear propagation, reflection and refraction, including Snell's law, with $n = c/v$. It also explains why waves *spread out* after passing through an opening, which is diffraction. The spreading is significant when the opening is comparable to the wavelength. That is why sound, with wavelengths of about a metre, bends readily around doors, while light, with wavelengths of about 500 nm, casts sharp shadows.

§23.2

Interference and coherence

When two waves overlap, their displacements add (superposition, Chapter 11). For light, the observable quantity is the intensity, $I \propto E^2$. Two waves of equal amplitude produce:

- **constructive interference** where they arrive **in phase**, with path difference $\Delta = m\lambda$, giving four times the intensity of one wave alone;
- **destructive interference** where they arrive **half a cycle out of phase**, with $\Delta = (m + \frac{1}{2})\lambda$, giving zero intensity.

A stable interference pattern needs **coherent** sources, which keep a fixed phase relationship. Light from ordinary sources is emitted as random bursts by atoms, with phases changing every nanosecond or so. Two separate bulbs therefore never produce visible fringes. The usual trick is to split light from a *single* source into two parts and then recombine them. Lasers are highly coherent, which makes interference easy to see.

§23.3

Young's double-slit experiment

In 1801 Thomas Young passed light from one source through two narrow, closely spaced slits, separated by d . On a distant screen he saw bright and dark bands, the decisive evidence that light is a wave. For a screen at a distance $L \gg d$, the path difference to a point at angle θ is $d \sin \theta$:

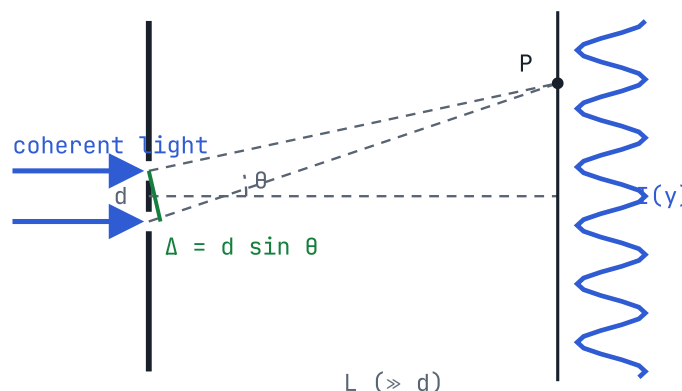
$$\text{bright fringes: } d \sin \theta = m\lambda, \quad \text{dark fringes: } d \sin \theta = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, \pm 1, \pm 2, \dots$$

For small angles, the position on the screen is $y = L \tan \theta \approx L \sin \theta$, so the bright fringes are at $y_m = m\lambda L/d$, with spacing

$$\Delta y = \frac{\lambda L}{d}.$$

The intensity pattern. With a phase difference $\phi = (2\pi/\lambda) d \sin \theta$ between the two waves:

$$I = I_{\max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right).$$



Young's double slit. The path difference $d \sin \theta$ decides whether the waves meet in phase (a bright fringe) or out of phase (a dark fringe) at the point P on the screen.

WORKED EXAMPLE 23.1

Measuring a wavelength

Two slits 0.20 mm apart are lit by a laser. On a screen 1.0 m away, the bright fringes are 3.0 mm apart. Find the wavelength. How far from the central bright fringe is the third bright fringe?

Solution

$$\lambda = \frac{d \Delta y}{L} = \frac{(2.0 \times 10^{-4})(3.0 \times 10^{-3})}{1.0} = 6.0 \times 10^{-7} \text{ m} = 600 \text{ nm (orange)}.$$

The third bright fringe is at $y_3 = 3\Delta y = 9.0 \text{ mm}$.

Evaluate. A wavelength of less than a micrometre is turned into a spacing of millimetres that you can measure with a ruler. Interference converts tiny lengths into big ones.

§23.4

Thin-film interference

Light reflected from the top and bottom surfaces of a thin film (a soap bubble, an oil slick, a lens coating) can interfere. Two factors decide the result.

1. **Extra path.** At near-normal incidence, the wave reflected from the bottom surface travels an extra distance $2t$ inside the film, where it has wavelength λ/n . This is equivalent to an extra path of $2nt$ in vacuum.
2. **Phase change on reflection.** A wave reflecting from a medium of *higher* index undergoes a phase change of π , equivalent to half a wavelength, just like a pulse reflecting from a fixed end of a string. Reflection from a *lower*-index medium gives no phase change.

KEY IDEA

Thin films, near-normal incidence.

- If exactly **one** of the two reflections has a phase change:
 - constructive when $2nt = \left(m + \frac{1}{2}\right)\lambda$;
 - destructive when $2nt = m\lambda$.
- If **both or neither** reflection has a phase change:
 - constructive when $2nt = m\lambda$;
 - destructive when $2nt = \left(m + \frac{1}{2}\right)\lambda$.

WORKED EXAMPLE 23.2

Soap films and anti-reflection coatings

(a) A soap film ($n = 1.33$) in air is 120 nm thick. Which visible wavelength is reflected most strongly? (b) A glass lens ($n = 1.52$) is coated with magnesium fluoride ($n = 1.38$) to minimise reflection at 550 nm. Find the minimum thickness of the coating.

Solution

(a) **Soap film.** There is a phase change at the top surface (air to soap) but not at the bottom (soap to air), so exactly one reflection has a phase change. Constructive interference needs

$$2nt = \left(m + \frac{1}{2}\right)\lambda \Rightarrow \lambda = \frac{2(1.33)(120)}{m + \frac{1}{2}} = \frac{319}{m + \frac{1}{2}} \text{ nm.}$$

With $m = 0$, $\lambda = 638$ nm. The film looks orange-red.

(b) **Coating.** Both reflections have a phase change (air to MgF_2 , and MgF_2 to glass), so destructive interference needs $2nt = \frac{1}{2}\lambda$:

$$t = \frac{\lambda}{4n} = \frac{550}{4(1.38)} = 99.6 \text{ nm.}$$

This is a “quarter-wave” coating.

Evaluate. As a soap film thins and drains, the colours shift. When it is much thinner than λ , the two reflected waves differ only by the π phase change, so they cancel for every colour, and the film looks *black* just before it bursts. Anti-reflection coatings on spectacles, camera lenses and solar cells cut the reflection from about 4% per surface to under 1%. Because they work best in the middle of the spectrum, they reflect a faint purple tint.

§23.5

The Michelson interferometer and OCT

A **Michelson interferometer** splits a beam in two with a half-silvered mirror, sends the two halves along perpendicular arms to mirrors, and recombines them. Moving one mirror by $\lambda/2$ changes that arm's path by λ , which shifts the fringe pattern by one whole fringe. This makes possible measurements of length to a small fraction of a wavelength. The LIGO detectors, Michelson interferometers with 4 km arms, detect gravitational waves that change the arm lengths by less than 10^{-18} m.

IN PRACTICE

Optical coherence tomography (OCT). OCT is a Michelson interferometer built for medicine. It uses a *broadband* near-infrared source, whose light is coherent only over a short **coherence length**, about $l_c \approx 0.44\lambda_0^2/\Delta\lambda$. One arm goes to a reference mirror, and the other into the patient's eye or skin. Interference appears **only** for light scattered back from the tissue layer whose path length matches the reference arm to within l_c . Scanning the reference (or analysing the spectrum) therefore maps reflectivity against depth. This builds micrometre-resolution cross-sections of the retina without touching it (Problem P23.15). OCT is now routine for diagnosing macular degeneration, diabetic eye disease and glaucoma, and catheter-based OCT images the walls of coronary arteries from the inside.

§23.6

Diffraction

Single-slit diffraction

Light passing through a single slit of width a spreads out into a central bright band flanked by weaker fringes. To find the dark fringes, divide the slit into two halves. At an angle where each wavelet in the top half is paired with one in the bottom half exactly $\lambda/2$ out of phase, every pair cancels. In general:

$$\text{dark fringes: } a \sin \theta = m\lambda, \quad m = \pm 1, \pm 2, \dots$$

Note that $m = 0$ is excluded: it is the bright centre.

The **central maximum** is twice as wide as the others, with angular half-width λ/a . On a screen at distance L , its full width is $2\lambda L/a$. The intensity is

$$I = I_0 \left(\frac{\sin \beta}{\beta} \right)^2, \quad \beta = \frac{\pi a \sin \theta}{\lambda}.$$

The narrower the slit, the wider the spread. This is the wave version of the uncertainty principle (Chapter 24).

WORKED EXAMPLE 23.3

A single slit

A He–Ne laser beam (633 nm) passes through a slit 0.10 mm wide onto a screen 2.0 m away. Find the width of the central bright band.

Solution

$$w = \frac{2\lambda L}{a} = \frac{2(633 \times 10^{-9})(2.0)}{1.0 \times 10^{-4}} = 2.5 \times 10^{-2} \text{ m} = 2.5 \text{ cm}.$$

A real double-slit pattern is the double-slit fringes multiplied by this single-slit envelope, because each slit has a finite width.

Circular apertures and the limit of resolution

A circular aperture of diameter D , such as a lens, a telescope mirror or the pupil of the eye, images a point source as a central bright disc (the **Airy disc**) surrounded by faint rings. The first dark ring is at

$$\sin \theta = 1.22 \frac{\lambda}{D}.$$

Two point sources are just **resolved** when the centre of one image falls on the first dark ring of the other. This is the **Rayleigh criterion**:

$$\theta_{\min} = 1.22 \frac{\lambda}{D}.$$

For a **microscope**, Ernst Abbe showed that the smallest resolvable separation is

$$d_{\min} \approx \frac{\lambda}{2 \text{NA}},$$

where $\text{NA} = n \sin \alpha$ is the objective's numerical aperture. Oil-immersion objectives reach $\text{NA} \approx 1.4$.

WORKED EXAMPLE 23.4**How sharp is your eye?**

Taking a pupil diameter of 3.0 mm and a wavelength of 550 nm, find the eye's diffraction-limited angular resolution. How far apart must two points 25 cm away be to be resolved? From how far away could you still see a car's two headlights (1.5 m apart) as separate?

Solution

$$\theta_{\min} = 1.22 \frac{550 \times 10^{-9}}{3.0 \times 10^{-3}} = 2.2 \times 10^{-4} \text{ rad } (\approx 0.8 \text{ arcmin}).$$

- At 25 cm: $s = (0.25)(2.2 \times 10^{-4}) = 56 \mu\text{m}$. - Headlights: $L = 1.5 / (2.2 \times 10^{-4}) \approx 6.7 \text{ km}$.

Evaluate. A healthy eye actually resolves about 1 arcminute. That is close to the diffraction limit, and it also matches the spacing of the cone cells in the fovea. Evolution has matched the “pixel size” to the optics. This is the basis of the “20/20” vision standard.

WORKED EXAMPLE 23.5**The resolution limit of the light microscope**

Find the smallest separation resolvable with an oil-immersion objective of NA 1.40, using green light at 500 nm. Can it resolve a bacterium (1 μm), a large virus (200 nm) and a small virus (30 nm)?

Solution

$$d_{\min} = \frac{500}{2(1.40)} = 180 \text{ nm}.$$

- The bacterium (1 μm) is clearly resolved. - The large virus (200 nm) is barely at the limit. - The small virus (30 nm) cannot be resolved.

Imaging viruses needs electron microscopy or super-resolution fluorescence methods.

Diffraction gratings

A **diffraction grating** has a very large number N of equally spaced slits or lines, a distance d apart. Bright maxima occur in the same directions as for two slits,

$$d \sin \theta = m\lambda,$$

but with many slits contributing, they become **extremely sharp**. A grating separates wavelengths far more precisely than a prism. Its **resolving power** is

$$R = \frac{\lambda}{\Delta\lambda} = Nm.$$

Gratings are the heart of spectrometers used in chemistry, in astronomy, and in clinical laboratory analysers.

WORKED EXAMPLE 23.6

A grating spectrometer

A grating has 600 lines per mm. (a) At what angles does the sodium line at 589 nm appear? (b) How many lines must be illuminated to resolve the sodium doublet (589.0 nm and 589.6 nm) in first order?

Solution

(a) $d = 1/600 \text{ mm} = 1.667 \mu\text{m}$.

- First order: $\sin \theta = 0.589/1.667 = 0.353$, so $\theta = 20.7^\circ$.
- Second order: $\sin \theta = 0.707$, so $\theta = 45.0^\circ$.
- Third order: $\sin \theta = 1.06 > 1$, which is impossible, so there is no third order.

(b) Resolving power:

$$R = \frac{589}{0.6} = 980 \Rightarrow N = \frac{R}{m} = 980 \text{ lines.}$$

So 980 lines, just 1.6 mm of this grating, are enough to resolve the doublet.

X-ray diffraction and Bragg's law

X-ray wavelengths (about 0.1 nm) are comparable to the spacing of atoms in crystals, so a crystal acts as a three-dimensional grating. X-rays reflect from planes of atoms a distance d apart, and the reflections reinforce only when

$$2d \sin \theta = m\lambda \quad (\text{Bragg's law}),$$

where θ is measured from the planes, not from the normal.

X-ray crystallography has determined the structures of salt, of drugs and enzymes, and, famously, of DNA. Rosalind Franklin's diffraction photograph 51 (1952) revealed its helical structure. Today it is central to structure-based drug design.

WORKED EXAMPLE 23.7**Bragg reflection from salt**

X-rays of wavelength 0.154 nm (copper K α) reflect from planes in rock salt that are 0.282 nm apart. Find the first-order Bragg angle.

Solution

$$\sin \theta = \frac{\lambda}{2d} = \frac{0.154}{0.564} = 0.273 \Rightarrow \theta = 15.8^\circ.$$

§23.7

Polarisation

Light is a transverse wave, so its electric field can oscillate in any direction perpendicular to the direction of travel. In **unpolarised** light, such as sunlight or the light from bulbs, the direction changes randomly. In **linearly polarised** light, $\vec{E}$ always oscillates along one direction.

Polarisers. A polarising filter transmits only the component of $\vec{E}$ along its transmission axis. Unpolarised light emerges with half its intensity. Polarised light of intensity I_0 passing through a polariser whose axis makes an angle θ with the light's polarisation obeys **Malus's law**:

$$I = I_0 \cos^2 \theta.$$

“Crossed” polarisers, at 90° , transmit nothing.

Polarisation by reflection. Light reflected from a surface is partly polarised parallel to the surface. At **Brewster's angle**,

$$\tan \theta_B = \frac{n_2}{n_1},$$

the reflected light is *completely* polarised. At this angle the reflected and refracted rays are perpendicular (Problem P23.18). For water, $\theta_B = 53^\circ$. Glare from roads, water and snow is therefore mostly horizontally polarised, and polarised sunglasses, whose transmission axis is vertical, block it.

Optical activity and birefringence. Some molecules, including sugars and many drugs, rotate the plane of polarisation by an amount proportional to their concentration and the path length.

Polarimeters use this to measure sugar concentrations, and to distinguish mirror-image drug molecules, which can have very different biological effects. Stressed plastics become *birefringent*:

their refractive index depends on the polarisation direction. Viewed between crossed polarisers they show coloured stress patterns, which engineers use to analyse stresses in models of structures and in orthopaedic implants.

WORKED EXAMPLE 23.8

Polarisers and Brewster's angle

(a) Unpolarised light of intensity I_0 passes through two polarisers whose axes are 60° apart. What intensity emerges? (b) Find Brewster's angle for light reflected from water.

Solution

(a) The first polariser halves the intensity, and the second applies Malus's law:

$$I = \frac{I_0}{2} \cos^2 60^\circ = \frac{I_0}{2} \cdot \frac{1}{4} = \frac{I_0}{8}.$$

(b) $\theta_B = \tan^{-1} 1.33 = 53^\circ$ from the normal.

CHAPTER SUMMARY

- Huygens' principle: every point on a wavefront is a source of wavelets. Diffraction is significant when an aperture is comparable to λ .
- Interference needs coherent sources. Double slit: bright fringes at $d \sin \theta = m\lambda$, spacing $\lambda L/d$, and $I \propto \cos^2(\pi d \sin \theta / \lambda)$.
- Thin films: path $2nt$, plus a π phase change on reflection from a higher-index medium. Quarter-wave anti-reflection coating: $t = \lambda/4n$.
- Single slit: dark fringes at $a \sin \theta = m\lambda$, with the central maximum of width $2\lambda L/a$.
- Resolution: $\theta_{\min} = 1.22\lambda/D$ (Rayleigh). Microscope: $d_{\min} \approx \lambda/(2NA)$.
- Grating: $d \sin \theta = m\lambda$, with resolving power $R = Nm$. Bragg: $2d \sin \theta = m\lambda$.
- Polarisation: unpolarised light is halved by a polariser. Malus: $I = I_0 \cos^2 \theta$. Brewster: $\tan \theta_B = n_2/n_1$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P23.1

Why don't two separate light bulbs shining on the same wall produce interference fringes, while two slits illuminated by the same laser do?

P23.2

A vertical soap film shows coloured bands, but its top appears black just before it bursts. Explain.

P23.3

Radio waves bend around buildings and hills, but light casts sharp shadows. Explain using the idea of diffraction.

P23.4

How do polarised sunglasses reduce glare from a wet road? In which direction should their transmission axis point? What would you see if you tilted your head by 90° ?

LEVEL B – STANDARD PROBLEMS

P23.5

In a double-slit experiment, the slits are 0.40 mm apart and the screen is 1.5 m away. The bright fringes are 2.5 mm apart. (a) Find the wavelength. (b) What would the spacing be if the whole apparatus were immersed in water ($n = 1.33$)?

P23.6

An oil film ($n = 1.45$) 300 nm thick floats on water ($n = 1.33$). Viewed from directly above in white light, what colour does it appear? (Find all wavelengths between 380 nm and 750 nm that are strongly reflected.)

P23.7

A camera lens ($n = 1.52$) is coated with a layer of MgF_2 ($n = 1.38$). (a) What is the minimum thickness that minimises reflection at 550 nm? (b) What is the next thickness that would also work? (c) Why does a coated lens look purplish?

P23.8

In single-slit diffraction with sodium light (589 nm), the first dark fringe appears at 1.50° from the centre. Find the width of the slit.

P23.9

The Hubble Space Telescope has a 2.4 m mirror. (a) Find its angular resolution at 500 nm. (b) Could it resolve a feature 10 m across on the Moon, which is 3.84×10^8 m away? What is the smallest lunar feature it can resolve?

P23.10

White light (400–700 nm) falls on a grating with 500 lines per mm. (a) Find the angular range of the first-order and second-order spectra. (b) Do the second and third orders overlap? Explain.

P23.11

Unpolarised light of intensity I_0 falls on two crossed polarisers, so that no light gets through. A third polariser is then inserted between them, at 45° to each. Find the transmitted intensity, and explain why adding a filter lets *more* light through.

P23.12

Polarimetry. Glucose rotates the plane of polarisation by 52.7° per decimetre of path per (g/mL) of concentration. A urine sample in a 1.0 dm tube rotates polarised light by 1.05° . Find the glucose concentration in g per 100 mL. Is it clinically significant? (Normal urine contains essentially no glucose.)

LEVEL C – CHALLENGE PROBLEMS**P23.13**

(a) Derive the double-slit intensity $I = I_{\max} \cos^2(\pi d \sin \theta / \lambda)$ by adding two waves of equal amplitude with a phase difference ϕ . (b) When the finite width a of each slit is included, the pattern is multiplied by the single-slit envelope $(\sin \beta / \beta)^2$. Show that if $d = 4a$, the 4th, 8th, ... bright fringes are missing. How many bright fringes then appear inside the central diffraction maximum?

P23.14

Air wedge. Two flat glass plates touch along one edge, and are separated at the other edge, 10 cm away, by a thin hair. Viewed from above in 600 nm light, 20 dark fringes are seen across the wedge, counting the one at the line of contact. (a) Explain why the line of contact is dark. (b) Find the thickness of the hair. (c) What is the fringe spacing?

P23.15

OCT resolution. An OCT system uses a source centred at $\lambda_0 = 840$ nm with a bandwidth $\Delta\lambda = 50$ nm. (a) Estimate its coherence length, $l_c \approx 0.44\lambda_0^2/\Delta\lambda$, which sets the axial (depth) resolution in air. (b) What is the resolution inside retinal tissue ($n \approx 1.38$)? (c) Why does a broader bandwidth give better depth resolution? (d) A Michelson mirror is moved by $0.25 \mu\text{m}$ while it is lit with 500 nm light. How many fringes pass a fixed point?

P23.16

Pixel density. A typical eye resolves about 1 arcminute (2.9×10^{-4} rad). (a) What is the smallest pixel size that is just invisible on a phone held 30 cm away? Express it as a number of pixels per inch (1 inch = 25.4 mm). (b) Repeat for a television viewed from 3.0 m. Comment on “retina display” marketing claims.

P23.17

X-rays and DNA. (a) In B-form DNA, the stacked base pairs are 0.34 nm apart. Find the first- and second-order Bragg angles for copper K α X-rays (0.154 nm) reflecting from these planes. (b) Explain why visible light could never reveal this spacing, and why X-ray wavelengths are ideal.

P23.18

Brewster’s angle. (a) Show that at Brewster’s angle the reflected and refracted rays are perpendicular, given that Brewster’s angle satisfies $\tan \theta_B = n_2/n_1$. (b) Find Brewster’s angle and the corresponding angle of refraction for light going from air into glass ($n = 1.50$). (c) Lasers often have “Brewster windows”, glass plates at Brewster’s angle at each end of the gas tube. Explain why this makes the laser’s output polarised.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are

listed in the References.

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